

Tushar Saurabh

Automation • Observability • Agentic AI

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PROFESSIONAL SUMMARY

I build software that automates repetitive work, simplifies troubleshooting, and improves engineering effectiveness. I apply agentic AI to automation and observability, building engineering workflows that reason over application telemetry, logs, code, and operational data while using deterministic validation where correctness and repeatability matter.

My work spans production troubleshooting, observability, engineering automation, code intelligence, AI-assisted software engineering, and application modernization. I remain hands-on while providing technical direction and mentoring engineers through real engineering problems.

SYSTEMS & ENGINEERING INITIATIVES

AI Software Engineering Harness

Designed and built a local-first multi-agent software engineering harness using Ollama, coordinating specialized agents for planning, implementation, review, testing, and verification. Built an independent execution and verification layer so generated implementations are validated against test cases rather than relying solely on model-reported outcomes.

Mainframe Modernization -- Ongoing

Building an application-discovery workflow that maps TELON screens and fields to underlying code and database relationships. Developed automation to exercise application flows, populate fields, and analyze resulting database changes, using runtime evidence to identify inconsistencies and validate relationships inferred during the initial analysis.

Application Intelligence Platform -- Ongoing

Building agent-assisted analysis of Dynatrace traces, logs, and server telemetry to examine failures, identify problem areas, score application performance and reliability, and detect sustained downward trends. The objective is to derive actionable engineering conclusions from observability data rather than relying solely on dashboards.

Code Intelligence

Developed and evolved code-intelligence approaches for understanding relationships across application repositories. Reworked an initially LLM-driven natural-language-to-Cypher approach after observing that differently worded versions of the same question could produce inconsistent queries, moving recurring user needs toward deterministic graph queries based on the information required by product owners, QA engineers, and developers.

Quantitative Investment Research Platform

Built an automated Python research pipeline for market-data ingestion, feature engineering, portfolio construction, optimization, backtesting, walk-forward validation, and performance analysis.

PROFESSIONAL EXPERIENCE

Chubb Business Services India

Senior Site Reliability Engineer | Jan 2024 -- Present

- Design and develop custom engineering applications that automate incident resolution and operational workflows, including tools that allow recurring incident actions to be completed in a few guided steps.
- Troubleshoot complex production issues using WinDbg, memory-dump analysis, application telemetry, logs, and server-level diagnostics to identify root causes that are not evident through conventional monitoring alone.
- Build custom application-performance and reliability workflows using Dynatrace, logs, traces, and server metrics to derive actionable insights rather than relying solely on pre-built dashboards.
- Build a pipeline of software-oriented SRE engineers through internship programs, defining real engineering problems across code intelligence, agentic UI, and AI-assisted observability, providing technical guidance, and continuing hands-on ownership and evolution of the resulting systems.
- Defined an agentic-UI research problem using a real application repository to investigate UI design characteristics that make browser agents inefficient or unreliable. Findings were shared with the development lead to inform

refactoring of existing UI solutions.

- Currently applying application-discovery and validation techniques to mainframe modernization, combining TELON screen/form analysis, code and database relationship discovery, and automated runtime validation.

Cognizant Technology Solutions (Client: Chubb)

Senior Software Engineer | Jul 2017 -- Jan 2024

- Built automation, observability solutions, and engineering tooling using Python, PowerShell, JavaScript, and modern web technologies for a large enterprise insurance application.
- Used AppDynamics, OpenSearch, and application analytics to investigate application behavior, identify performance bottlenecks, and improve operational visibility.
- Collaborated with DevOps teams on CI/CD workflows and used Jenkins and Ansible as part of application deployment and operational automation.

Cognizant Technology Solutions (Client: Chubb)

Senior Associate | Apr 2010 -- Jul 2017

Worked on enterprise insurance applications on IBM Mainframe and created automation workflows to reduce repetitive engineering work. In 2013, learned Python and moved into a custom web application, progressively expanding into web technologies and automation. This transition established the foundation for the modern application, production engineering, and observability work that followed.

Infosys

Technology Analyst | Jun 2006 -- Apr 2010

Developed and supported enterprise mainframe applications, with COBOL and DB2 as the primary technologies. Created automation workflows to reduce repetitive development and operational tasks. Received the Infosys Excellence Award and Prima Award for contributions to project execution and delivery.

TECHNICAL LEADERSHIP

- Define engineering problem statements for internship programs and mentor interns while allowing them to independently investigate and develop solutions.
- Use internships as a long-term strategy for developing SRE engineers with stronger software engineering and automation capabilities; members of the first internship cohort subsequently joined the team.
- Continue to maintain, modify, and evolve the engineering systems after internship projects conclude, including making architectural changes when experimental approaches prove inconsistent or unsuitable.
- Experience coaching engineers while continuing to pursue a hands-on individual-contributor and technical-leadership path.

ENGINEERING TOOLKIT

Programming: Python, PowerShell, JavaScript, TypeScript, SQL

Observability & Troubleshooting: Dynatrace, AppDynamics, OpenSearch, WinDbg, Sysinternals

Engineering & Automation: Git, GitHub, Jenkins, Ansible, Playwright

AI Engineering: Claude Code, Ollama, OpenAI API, Agentic workflows, Multi-agent orchestration

Application Development & Data: Node.js, Angular, Svelte, Pandas, NumPy

Systems: Windows, Linux, Mainframe application analysis, Production diagnostics

EDUCATION

University of Pennsylvania

Master of Computer and Information Technology | 2021 -- 2024

International Institute of Information Technology Hyderabad (IIIT-H)

Master of Science in Data Science (MSc DS) -- In Progress

AWARDS

Infosys Excellence Award • Prima Award -- Project Execution